

Assignment No 04

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Question:

Friend function and Friend Class

Code:

1. Student

```
#include <iostream>
#include <string>
#include <limits>
using namespace std;

class Student {
private:
    int roll;
    string name;
    float marks[5];
    float percentage;
    string division;

    int validRoll() {
        int x;
        while (true) {
            cin >> x;
            if (cin.fail() || x <= 0) {
                cout << "Enter valid roll: ";
                cin.clear();
                cin.ignore(numeric_limits<streamsize>::max(), '\n');
            } else return x;
        }
    }

    string validName() {
        string n;
        while (true) {
            getline(cin, n);
            bool ok = true;
            if (n.empty()) ok = false;
            for (char c : n) {
                if (!(isalpha(c) || c == ' ')) {
                    ok = false;
                    break;
                }
            }
            if (ok) return n;
            cout << "Enter valid name: ";
        }
    }
}
```

```

}

float validMarks() {
    float m;
    while (true) {
        cin >> m;
        if (cin.fail() || m < 0 || m > 100) {
            cout << "Enter marks 0-100: ";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        } else return m;
    }
}

public:
    Student() {
        roll = 0;
        name = "";
        for (int i = 0; i < 5; i++) marks[i] = 0;
        percentage = 0;
        division = "";
    }

    ~Student() {}

    void input() {
        cout << "Enter roll: ";
        roll = validRoll();

        cout << "Enter name: ";
        cin.ignore();
        name = validName();

        for (int i = 0; i < 5; i++) {
            cout << "Enter marks " << i + 1 << ": ";
            marks[i] = validMarks();
        }
    }

    void calculatePercentage() {
        float sum = 0;
        for (int i = 0; i < 5; i++) sum += marks[i];
        percentage = sum / 5.0;
    }

    void determineDivision() {
        if (percentage >= 60) division = "First";
        else if (percentage >= 50) division = "Second";
        else if (percentage >= 40) division = "Third";
        else division = "Fail";
    }

    void display() {
        cout << "Roll: " << roll << endl;
    }

```

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        cout << "Name: " << name << endl;
        cout << "Marks: ";
        for (int i = 0; i < 5; i++) cout << marks[i] << " ";
        cout << endl;
        cout << "Percentage: " << percentage << endl;
        cout << "Division: " << division << endl;
    }

    friend void comparePercentage(Student s1, Student s2);
};

void comparePercentage(Student s1, Student s2) {
    if (s1.percentage > s2.percentage) {
        cout << "Higher Percentage Student:\n";
        s1.display();
    } else if (s2.percentage > s1.percentage) {
        cout << "Higher Percentage Student:\n";
        s2.display();
    } else {
        cout << "Both have same percentage\n";
        s1.display();
        s2.display();
    }
}

int main() {
    Student s1, s2;

    cout << "Enter details of student 1\n";
    s1.input();
    s1.calculatePercentage();
    s1.determineDivision();

    cout << "Enter details of student 2\n";
    s2.input();
    s2.calculatePercentage();
    s2.determineDivision();

    comparePercentage(s1, s2);

    return 0;
}

```

Output:

```
(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 4> .\student.exe
Enter details of student 1
Enter roll: 28
Enter name: om
Enter marks 1: 69
Enter marks 2: 57
Enter marks 3: 29
Enter marks 4: 38
Enter marks 5: 28
Enter details of student 2
Enter roll: 36
Enter name: 97
Enter valid name: 92
Enter valid name: 83
Enter valid name: 11
Enter valid name: tommy
Enter marks 1: 2
Enter marks 2: 697
Enter marks 0-100: 13
Enter marks 3: 58
Enter marks 4: 69
Enter marks 5: 12
Higher Percentage Student:
Roll: 28
Name: om
Marks: 69 57 29 38 28
Percentage: 44.2
Division: Third
❖(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 4> █
```

2. Bank:

```
#include<iostream>
#include<limits>
using namespace std;

class Accounts;

class Bank{
public:
    int minBalance;
    Accounts* acc;
    int total;

    Bank();
};

class Manager;

class Accounts{
    friend class Manager;
public:
    float balance;
    int PIN;
    string accountHolder;
    static int accountID;
    Bank* bank;
    int id;

    float validAmount(){
        float x;
        while(true){
            cin>>x;
```

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        if(cin.fail() || x < 0){
            cout<<"Invalid amount. Re-enter: ";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }else return x;
    }
}

string validName(){
    string name;
    while(true){
        getline(cin, name);

        bool ok = true;
        if(name.empty()){
            ok = false;
        }

        for(char c : name){
            if(!(isalpha(c) || c==' ')){
                ok = false;
                break;
            }
        }

        if(ok) return name;
        cout<<"Invalid name. Only alphabets allowed. Re-enter: ";
    }
}

int validPIN(){
    int p;
    while(true){
        cin>>p;
        if(cin.fail() || p<1000 || p>9999){
            cout<<"Enter 4-digit PIN: ";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }
        else{
            return p;
        }
    }
}

int getValidID(Bank &bank){
    int id;
    while(true){
        cout<<"Enter account ID: ";
        cin>>id;

        if(cin.fail()){
            cout<<"Invalid ID. Enter numeric value.\n";
            cin.clear();

```

```

        cin.ignore(numeric_limits<streamsize>::max(), '\n');
        continue;
    }

    for(int i=0;i<bank.total;i++){
        if(bank.acc[i].id == id)
            return id;
    }

    cout<<"Account not found. Try again.\n";
}

}

void createAccount(string name, float initialBalance, int pin){
    id = accountID++;

    cout<<"Enter Name: ";
    cin.ignore();
    accountHolder = validName();

    cout<<"Enter Initial Balance (>= "<<bank->minBalance<<"): ";
    while(true){
        balance = validAmount();
        if(balance < bank->minBalance)
            cout<<"Minimum balance required. Re-enter: ";
        else break;
    }

    cout<<"Set PIN: ";
    PIN = validPIN();

    cout<<"Account created successfully! ID: "<<id<<endl;
}

bool verify(){
    int p;
    cout<<"Enter PIN: ";
    cin>>p;
    return p == PIN;
}

void viewBalance(){
    if(verify())
        cout<<"\nBalance: "<<balance<<endl;
    else
        cout<<"Wrong PIN\n";
}

void withdraw(){
    if(!verify()){
        cout<<"Wrong PIN\n";
        return;
    }
}

```

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        cout<<"Enter amount: ";
        float amt = validAmount();

        if(balance - amt < bank->minBalance){
            cout<<"Cannot go below minimum balance\n";
            return;
        }

        balance -= amt;
        cout<<"Withdraw successful\n";
    }

    void resetPIN(){
        if(!verify()){
            cout<<"Wrong PIN\n";
            return;
        }

        cout<<"Enter new PIN: ";
        PIN = validPIN();
        cout<<"PIN updated\n";
    }
};

Bank::Bank(){
    minBalance = 500;
    acc = new Accounts[10];
    total = 0;
}

class Manager{
private:
    int managerPIN = 9999;

    bool authenticate(){
        int p;
        cout<<"Enter Manager PIN: ";
        cin>>p;
        if(p == managerPIN){
            return true;
        }
        cout<<"Wrong Manager PIN\n";
        return false;
    }

public:
    void viewAll(Bank &b){
        if(!authenticate()) return;

        cout<<endl<<"-----MANAGER-----"<<endl;
        for(int i=0;i<b.total;i++){
            cout<<"ID: "<<b.acc[i].id<<endl;
            cout<<"Name: "<<b.acc[i].accountHolder<<endl;
            cout<<"Balance: "<<b.acc[i].balance<<endl;
        }
    }
};

```

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    }
}

};

int Accounts::accountID = 1;

int main(){
    int flag = 1;
    Bank bank;
    Manager m;

    do{
        cout<<"\n\n-----"<<endl;
        cout<<"1. Create Account"<<endl;
        cout<<"2. View Balance"<<endl;
        cout<<"3. Withdraw"<<endl;
        cout<<"4. Reset PIN"<<endl;
        cout<<"5. Manager Priviledges"<<endl;
        cout<<"0. Exit"<<endl;
        cout<<"-----"<<endl;
        cout<<"Enter your choice: ";
        cin>>flag;

        switch(flag){
            case 1:{
                if(bank.total>=10){
                    cout<<"Bank full\n";
                    break;
                }
                bank.acc[bank.total].bank = &bank;
                bank.acc[bank.total].createAccount("",0,0);
                bank.total++;
                break;
            }

            case 2:{
                if(bank.total==0){
                    cout<<"No accounts exist\n";
                    break;
                }
                int id = bank.acc[0].getValidID(bank);
                for(int i=0;i<bank.total;i++)
                    if(bank.acc[i].id==id)
                        bank.acc[i].viewBalance();
                break;
            }

            case 3:{
                if(bank.total==0){
                    cout<<"No accounts exist\n";
                    break;
                }
                int id = bank.acc[0].getValidID(bank);
                for(int i=0;i<bank.total;i++)

```



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        if(bank.acc[i].id==id)
            bank.acc[i].withdraw();
        break;
    }

    case 4:{
        if(bank.total==0){
            cout<<"No accounts exist\n";
            break;
        }
        int id = bank.acc[0].getValidID(bank);
        for(int i=0;i<bank.total;i++)
            if(bank.acc[i].id==id)
                bank.acc[i].resetPIN();
        break;
    }

    case 5:
        m.viewAll(bank);
        break;

    case 0:
        cout<<"Exiting..."<<endl;
        break;

    default:
        cout<<"Invalid choice. Please try again."<<endl;
    }

}while(flag != 0);

return 0;
}

```

Output:

```

(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 4> .\main.exe

-----
1. Create Account
2. View Balance
3. Withdraw
4. Reset PIN
5. Manager Priviledges
0. Exit
-----
Enter your choice: 1
Enter Name: 12
Invalid name. Only alphabets allowed. Re-enter: om
Enter Initial Balance (>= 500): 500
Set PIN: 1234
Account created successfully! ID: 1

-----
1. Create Account
2. View Balance
3. Withdraw
4. Reset PIN
5. Manager Priviledges
0. Exit
-----
Enter your choice: 2
Enter account ID: 1
Enter PIN: 1234

Balance: 500

-----
1. Create Account
2. View Balance
3. Withdraw
4. Reset PIN
5. Manager Priviledges
0. Exit
-----

```

```

-----
Enter your choice: 3
Enter account ID: 1
Enter PIN: 1234
Enter amount: 200
Cannot go below minimum balance

-----
1. Create Account
2. View Balance
3. Withdraw
4. Reset PIN
5. Manager Priviledges
0. Exit
-----
Enter your choice: 5
Enter Manager PIN: 9999

-----MANAGER-----
ID: 1 | Name: om | Balance: 500

-----
1. Create Account
2. View Balance
3. Withdraw
4. Reset PIN
5. Manager Priviledges
0. Exit
-----
Enter your choice: 0
Exiting...
(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 4>

```